INCH-POUND

MIL-DTL-32642
01 July 2020

DETAIL SPECIFICATION

CAP, BLASTING: NON-ELECTRIC, 5 MINUTE TIME FUSE DELAY, M14, LOAD, ASSEMBLE, AND PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements and verification methods for the following items:

CAP, BLASTING: NON-ELECTRIC, 5 MINUTE TIME FUSE DELAY, M14

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army CCDC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERICAL ITEM DESCRIPTIONS

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-59383	-	Marking Media, Opaque (Porous and Non-Porous Surfaces)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32411	-	Igniter, Time Blasting, Fuse: With Shock Tube Capability – M81, Load, Assemble, and Pack
MIL-DTL-45144	-	Fuse, Blasting, Time- M700, and Dummy Fuse, Blasting, Time - Inert Loading, Assembly and Packing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-650	-	Explosive; Sampling, Inspection and Testing
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Products
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US ARMY CCDC - AC DRAWINGS

8830953	-	Cup
9325874	-	Packing for Light Boxes
12972620	-	Igniter, Time, Blasting Fuse: With Shock Tube Capability M81
12988007	-	Packing and Marking for Cap, Blasting: Non-Electric, 5 Minute Time Fuse Delay, M14 (10 Units/Fiberboard Box, 4 Fiberboard Boxes/Wood Box, 40 Total Units/Wood Box)

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13023563	-	Combination of Adopted Items for: Cap, Blasting: Non-Electric, 5 Minute Time Fuse Delay, M14 (10 units/pack, 4 packs/wood box, 40 total units/wood box)
13026127	-	Blasting Cap Assembly
13028635	-	Cap, Blasting: Non-Electric, 5 Minute Time Fuse Delay, M14

(Copies of the above drawings are available by email at usarmy.pica.ccdc-ac.list.drawing-request-help-desk@mail.mil.)

US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116/121U	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Cap, Blasting, Non-Electric, Delay, M14, Packed 40 Per Wooden Box, Unitized 24 Boxes Per 40"X 48" Pallet; Approx Box Size 25-7/8'L X 13-3/4"W X 12-1/4" H

(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Default.aspx.)

INTERNATIONAL TEST OPERATING PROCEDURES

ITOP 1-2-601	-	Laboratory Vibration Schedules, International Test Operations Procedure

(Copies of the above international test procedures are available online at http://www.atec.army.mil/publications/topsindex.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A108	-	Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished

(Copies of the above documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Materials, components, and assemblies. All materials, components and assemblies shall comply with the requirements of Drawing 13028635 and its subsidiary drawings.

3.3 Blasting cap attachment integrity. The blasting cap and time fuse shall not separate and shall function after exposure to an axial force not less than fifteen pounds (15 lbs) for not less than five (5) seconds.

3.4 Tensile integrity. The time fuse shall not exhibit evidence of damage and shall function after exposure to an axial force not less than twenty five pounds (25 lbs) for not less than five (5) seconds.

3.5 Initiation. The M14 shall initiate upon receipt of a fire signal from an M81 Igniter, Drawing 12972620, shall have a burn time in accordance with Drawing 13028635 (see 6.3), and the blasting cap shall function high order (see 6.4).

3.6 Operating temperatures. The M14 shall operate throughout the temperature range of -50˚F to 130˚F.

3.7 Storage temperatures. The M14 shall not exhibit evidence of damage and shall function without performance degradation after being subjected to temperature extremes at -65˚F and 165˚F.

3.8 Transportation/vibration. The M14 shall not exhibit evidence of damage and shall function without performance degradation after being subjected to transportation vibration at a temperature range of -65˚F to 165˚F.

3.9 Functioning output. After detonation of the blasting cap, the steel block shall have a dent depth of not less than 0.010 inches.

3.10 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.11 Workmanship. All parts, sub-assemblies, and assemblies shall be clean and free of burrs, chips, sharp edges, cracks, unblended radii, burn marks, checks, sinks, crazes, blisters, excess flash, dirt, grease, oil, rust, corrosion, and other foreign matter.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 3 - Examination 2 - Demonstration 4 - Test		Classes of Verification A – First article B – Conformance						
Section 3 and 5 Requirements	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article		X	X	X	X		4.2
3.1.2	Conformance inspection		X	X	X		X	4.3
3.2	Materials, components, and assemblies			X	X	X	X	4.3.4.1 – 4.3.4.4
3.3	Blasting cap attachment integrity				X	X	X	4.5.2
3.4	Tensile integrity				X	X	X	4.5.3
3.5	Initiation				X	X	X	4.4.1
3.6	Operating temperatures				X	X	X	4.5.4
3.7	Storage temperatures				X	X	X	4.5.5
3.8	Transportation/vibration				X	X		4.5.6
3.9	Functioning output				X	X		4.5.7
3.10	Ammunition lot numbering			X		X	X	4.3.3
3.11	Workmanship			X		X	X	4.3.4.1 - 4.3.4.11
5.1	Packaging and marking			X	X	X	X	4.3.4.5 - 4.3.4.11, 4.6.4
5.1.1	Barrier bag resistance to leakage				X	X	X	4.6.2, 4.6.3
5.1.2	Marking ink/media			X		X	X	4.3.4.5 - 4.3.4.11

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article inspection in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures and facilities as will be used during regular production.

4.2.1 First article rejection. If one (1) or more assemblies, subassemblies, materials, or components fail to comply with the first article requirements, the sample(s) and first article shall be rejected.

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TABLE II. First article inspections and tests.

Examination or test	Drawing(s)	Sample Size	Requirement Paragraph	Paragraph Reference / Inspection Method
Cup	8830953-2	30	3.2	4.3.4.1
Cap, blasting, non-electric	13026127-2	30	3.2	4.3.4.2
Fuse, blasting, time	13064970	MIL-DTL-45144	3.2	MIL-DTL-45144
M14 assembly	13028635			
Examination for defects 1/		30	3.2	4.3.4.3
Blasting cap attachment integrity		30	3.3	4.5.2
Tensile integrity		30	3.4	4.5.3
Functioning output		30	3.5, 3.9	4.5.7
Storage temperatures 1/	12988007	6 fiberboard boxes	3.5, 3.7	4.5.5
Operating temperatures		60	3.6	4.5.4
Hot		20	3.5, 3.6	4.5.4.a
Ambient		20	3.5, 3.6	4.5.4.b
Cold		20	3.5, 3.6	4.5.4.c
Transportation/vibration 1/	12988007	2 wood boxes	3.8	4.5.6
Hot		40	3.5, 3.8	4.5.6.a
Cold		40	3.5, 3.8	4.5.6.b
Packaging and marking	12988007	2/	5.1	4.3.4.5 - 4.3.4.11
Fiberboard box (inner)		2/	5.1	4.3.4.5 - 4.3.4.6
Barrier Bags		2/	5.1	4.3.4.7
Leakage test		3/	5.1.1	4.6.2
Heat seal seam strength test		3/	5.1.1	4.6.3
Fiberboard box (outer)		2/	5.1	4.3.4.8
Wood box		2/	5.1	4.3.4.9 - 4.3.4.10
Unitized load	19-48-4116/121U	2/	5.1	4.3.4.11

Notes:
1/ Indentation represents sequential testing. The sequence shall follow in the order listed.
2/ For packaging: The examinations shall be performed on one (1) full pack and one (1) light pack.
For light packs: The light pack shall be no less than 50% filled and the remainder packed out per Drawing 9325874.
For unitized loads: The examinations shall be performed on one (1) sample.
Packaging within unitized load or assembled packaging may include samples from lower level packaging when conformance can still be demonstrated at all levels. With prior Government approval, the samples to be enclosed in the packaging may be substituted with inert mass and volume analogues, reused for each sample, or excluded from packaging if not necessary to demonstrate conformance.
3/ The test shall be performed on three (3) sealed barrier bags from each sealer.

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4.3 Conformance inspection.

4.3.1 Conformance rejection. If any assembly, subassembly, material, or component fails to comply with the conformance inspection requirements of 4.3 and its subparagraphs, the lot from which the sample was taken shall be rejected.

4.3.2 Inspection lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. Inspection lots shall contain materials from not more than one lot interfix number from one manufacturer (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specifications and revisions. In addition each lot shall contain materials from not more than one lot interfix number from one manufacturer except the following:

a. RDX from not more than two lots from one lot interfix number from one manufacturer.
b. Lead azide from not more than two lots from one lot interfix number from one manufacturer.
c. Lead styphnate from not more than two lots from one lot interfix number from one manufacturer.
d. M700 from not more than one lot from one lot interfix number from one manufacturer.

4.3.3 Ammunition lot number. Visually verify that a unique ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.3.4 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.11, the definitions of critical, major and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph “Acceptance by contractor-proposed provisions” of MIL-STD-1916).

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4.3.4.1 Cup.			Drawing Number 8830953-2 Next Higher Assembly 13026127-2	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Cup split or cracked	VL IV	3.2	Visual
102	Outside diameter	VL IV	3.2	Gage
103	Concavity of bottom	VL IV	3.2	Gage
104	Thickness of bottom	VL IV	3.2	Gage
Minor				
201	Inside diameters	VL II	3.2	Gage
202	Workmanship	VL II	3.11	Visual

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4.3.4.2 Cap, blasting, non-electric.			Drawing Number 13026127-2 Next Higher Assembly 13028635	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Moisture content of lead azide (or alternate intermediate and ignition charge)	1/	3.2	4.5.1
102	Total RDX weight	VL IV	3.2	Gage
103	Lead azide weight	VL IV	3.2	Gage
104	Lead styphnate weight	VL IV	3.2	Gage
105	Total alternate intermediate and ignition charge weight (when applicable) 2/	VL IV	3.2	Gage
106	Cup split or cracked	VL IV	3.2	Visual
Minor				
201	Workmanship	VL II	3.11	Visual

Note:
1/ Samples shall be taken from the quantity of lead azide used in each shift of lead azide loading. For sampling purposes, each daily shift shall be not greater than 10 hours.
2/ When alternate intermediate and ignition charge is used, Majors 103 and 104 shall not apply.

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4.3.4.3 Cap, blasting: non-electric, 5 minute time fuse delay, M14 - examinations.			Drawing Number 13028635 Next Higher Assembly 12988007	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical				
1	Cup split or cracked, exposed charge, or evidence of explosive leakage	100% + VL VII	3.2	Visual
Major				
101	Location of 1st crimp from blasting cap edge	VL IV	3.2	Gage
102	Location of 2nd crimp from 1st crimp	VL IV	3.2	Gage
103	Crimp widths	VL IV	3.2	Gage
104	Crimp diameters	VL IV	3.2	Gage
105	Time fuse color incorrect	VL IV	3.2	Visual
106	Time fuse not a continuous strand (contains splices)	VL IV	3.2	Visual
107	Minute marking quantity or color incorrect	VL IV	3.2	Visual
108	Time fuse surface defects including cracks, holes, bubbles and breaks	VL IV	3.2	Visual
109	Protective cap not present or not fully installed	VL IV	3.2	Visual / Manual
110	Length from 5 minute mark to start of protective cap	VL IV	3.2	Gage
Minor				
201	Workmanship	VL II	3.11	Visual

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4.3.4.4 Cap, Blasting: Non-Electric, 5 Minute Time Fuse Delay, M14 - destructive.			Drawing Number 13028635 Next Higher Assembly 12988007	
	Examinations and Tests	Conformance Criteria 1/	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical				
1	Burn time 50% less than desired time increment	100%	3.2	4.4.1
Major				
101	Blasting cap attachment integrity	30	3.3	4.5.2
102	Tensile integrity	30	3.4	4.5.3
103	Functioning output	30	3.9	4.5.7
104	Hot operating temperature	10	3.6	4.5.4.a
105	Ambient operating temperature	10	3.6	4.5.4.b
106	Cold operating temperature	10	3.6	4.5.4.c
Minor	None defined			

Notes:
1/ Perform all the examinations of 4.3.4.3 on samples prior to testing.
The total number of samples is sixty (60).
Accept on zero (0) defects; reject on one (1) or more defects.
The order of sequential testing shall be Majors 101, 102, then 103.

4.3.4.5 Packing and marking drawing for the fiberboard box (inner, unsealed).			Drawing Number 12988007 Next Higher Assembly 19-48-4116/121U	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	VL II	5.1	Visual
102	Packaging material moisture content incorrect	4.6.1	5.1	4.6.1
103	Blasting cap(s) falls out when turned upside down	VL II	5.1	Manual
Minor				
201	Workmanship	VL I	3.11	Visual

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4.3.4.6 Packing and marking drawing for the fiberboard box (inner, sealed).			Drawing Number 12988007 Next Higher Assembly 19-48-4116/121U	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Sealing strip torn, badly wrinkled or otherwise fails to seal box completely or correctly	VL II	5.1	Visual
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	Visual
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.11	Visual

4.3.4.7 Packing and marking drawing for the barrier bag (sealed).			Drawing Number 12988007 Next Higher Assembly 19-48-4116/121U	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	Visual
102	Leakage test	1/	5.1.1	4.6.2
103	Heat seal seam strength test	1/	5.1.1	4.6.3
Minor				
201	Workmanship	VL I	3.11	Visual
Note: 1/ The test shall be performed at the beginning of each shift on one (1) sealed barrier bag from each sealer. For sampling purposes, consider a shift to be no be more than ten (10) hours.				

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4.3.4.8 Packing and marking drawing for the fiberboard box (sealed).			Drawing Number 12988007 Next Higher Assembly 19-48-4116/121U	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Sealing strip torn, badly wrinkled or otherwise fails to seal box completely or correctly	VL II	5.1	Visual
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	Visual
Minor				
201	Contents loose	VL I	5.1	Visual
202	Workmanship	VL I	3.11	Manual

4.3.4.9 Packing and marking drawing for the wood box (unsealed).			Drawing Number 12988007 Next Higher Assembly 19-48-4116/121U	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	VL II	5.1	Visual
Minor				
201	Workmanship	VL I	3.11	Visual

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4.3.4.10 Packing and marking drawing for the wood box (sealed).			Drawing Number 12988007 Next Higher Assembly 19-48-4116/121U	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Box, hinge, or hasp damaged	VL II	5.1	Visual
102	Metallic seal(s) missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	VL II	5.1	4.6.4
103	Markings –including 2D bar codes along with wood preservative and heat treatment markings - missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	Visual / Gage
Minor				
201	Contents loose	VL I	5.1	Visual
202	Workmanship	VL I	3.11	Visual

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4.3.4.11 Unitization.			Drawing Number 19-48-4116/121U Next Higher Assembly	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	VL II	5.1	Visual
102	Pallet load markings -including 2D bar codes- missing, incorrect, illegible, or not in proper location	VL II	5.1.2	Visual
103	Filler assembly not built, treated, or marked correctly	VL II	5.1	Visual
104	Boxes incorrectly oriented or positioned on pallet	VL II	5.1	Visual
105	Strapping not applied correctly or incorrect type	VL II	5.1	Visual
106	Unit load not tightly strapped or any strap loose	VL II	5.1	Visual
Minor				
201	Workmanship	VL I	3.11	Visual

4.4 Test sub-procedures.

4.4.1 Functioning procedure. Each lot of the M81 igniter shall be in accordance with MIL-DTL-32411, except for the marking and packaging requirements. Obtain an M81 and loosen the fuse holder cap three revolutions. Loosen the plugs by pressing them into the igniter. Grip both the weatherproofing and shipping plugs and move them side to side while simultaneously pulling both plugs from the igniter. On the M14 sample, locate the five (5) minute timing mark (next to the protective cap). Cut squarely in the approximate center of the mark. Insert the fuse into the igniter as far as it will go (to ensure complete and proper insertion of the fuse). Re-tighten the fuse holder cap to ensure proper holding force; fingertight closures shall be used. Remove the cotter pin. Apply an axial force to the pull ring. If the M81 does not function, reset per 4.4.1.1. Record the burn time from the M81 initiation point to the blasting cap detonation point and verify that the burn time is not less than 300 seconds and not greater than 345 seconds (see Table III and 6.3). Verify the complete detonation of the sample's blasting cap (see 6.4). Record the barometric pressure, humidity, and temperature of the test area. The

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sample(s) and lot shall be rejected if the burn time fails to meet the requirement or the blasting cap does not function high order.

4.4.1.1. Quick reset procedures. Reengage the striker assembly by pushing the pull rod back into the end of the igniter. Verify the pull rod re-engages the striker by feeling the pull ring's increased resistance to withdrawal. After the pull rod re-engages the striker, apply an axial force to the pull ring. If the igniter does not function after the first reset, it may be reset a second time. If the M81 igniter is incapable of being reset or if it is reset twice and still does not function, replace the M81 igniter. The replacement igniter and its lot number shall be recorded.

4.5 Test methods.

4.5.1 Moisture content of lead azide (alternate mixture). The moisture content of the lead azide (or alternate intermediate and ignition charge, when applicable) at the time of loading shall be determined in accordance with Method 101.5 of MIL-STD-650, except that a 2.0 gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene. To ensure that the lead azide (alternate mixture) complies with the requirement, adequate controls shall be in place. Composite samples shall not be permitted. If the moisture content of a sample exceeds the requirement and loading has not begun, the lead azide (alternate mixture) represented by the sample shall be rejected. If the moisture content of a sample exceeds the requirement and components have been loaded with lead azide (alternate mixture) containing excessive moisture, the loaded components and the remaining unloaded lead azide (alternate mixture) shall be rejected.

4.5.2 Blasting cap attachment integrity. Secure the blasting cap. Apply the specified axial load to the junction of the blasting cap and time fuse the specified length of time. Verify that the blasting cap remains attached to the time fuse. The sample(s) and lot shall be rejected if there is separation of the blasting cap. Upon successful completion, subject the sample(s) to 4.5.3.

4.5.3 Tensile integrity. Secure a portion of time fuse not less than twenty four (24) inches from the blasting cap and not less than twelve (12) inches in length. Apply the specified axial force to the time fuse for the specified length of time. The sample(s) and lot shall be rejected if any of the time fuse separates. Upon successful completion, subject the sample(s) to 4.5.7.

4.5.4 Operating temperatures. Expose the sample(s) of M14 to the following temperatures for not less than four (4) hours:

a.	Hot	125°F +5/-5°F	(see 4.5.4.1)
b.	Ambient	75°F +5/-5°F	
c.	Cold	-45°F +5/-5°F	

At not greater than fifteen (15) minutes of the removal of the conditioned sample(s), the sample(s) shall be inspected for compliance with the workmanship requirement of 3.11 and

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initiated in accordance with 4.4.1. The sample(s) and lot shall be rejected if the M14 does not remain in compliance with the workmanship or functioning requirement.

4.5.4.1. Operating temperature – hot burn time. In accordance with Table III, count the required number of marking increments from the blasting cap and cut squarely in the approximate center of the mark. Initiate the sample in accordance with 4.4.1. Verify compliance with the Table III burn time requirement.

TABLE III. Hot temperature burn requirements.

Number of samples		Minute increment to be tested	Adjusted burn time requirement (seconds) (see 6.3)
FAT	LAT		
10	5	5	300 +45/-0
10	5	1	60 +9/-0

4.5.5 Storage temperatures. The test shall be conducted on the M14's in their fiberboard box. Temperature condition for not less than seventy two (72) hours at the following temperatures:

a. Hot 160°F +5/-5°F
b. Cold -60°F +5/-5°F

After temperature conditioning, the sample(s) shall be unpackaged and inspected for compliance with the workmanship requirement of 3.11. The sample(s) and lot shall be rejected if the M14 does not remain in compliance with the workmanship requirement.

4.5.6 Transportation/vibration. Perform the transportation/vibration test in accordance with ITOP 1-2-601 on the M14 samples in their wood box(es). Follow ITOP 1-2-601's test procedure of 4.1.1 "Testing of Ammunition." Condition the wood box(es) for not less than twenty four (24) hours to the following temperatures:

a. Hot 160°F +5/-5°F
b. Cold -60°F +5/-5°F

After conditioning, maintain the temperature and conduct the vibration according to the "Wheeled Vehicles" and the "Two-wheeled Trailers" schedules. After not greater than fifteen (15) minutes of the removal of the conditioned samples, each sample shall be inspected for compliance with 3.11 and initiated in accordance with 4.4.1 at room temperature. The sample(s) and lot shall be rejected if the M14 does not remain in compliance with the workmanship or functioning requirement.

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4.5.7 Functioning output. Perform output testing per MIL-STD-331 Test D4 (Dent Block Method) on the M14. The dent block shall be made of grade 1018 steel per ASTM A108, have a Rockwell Hardness between B70-B95, and be 1.250 x 1.250 x 0.625 +/- 0.1 inches in size. Secure each blasting cap perpendicularly to a dent block using a test fixture, which only contacts the donor's blasting cap where there is no RDX, Lead Azide, or Lead Styphate and also does not create any confinement on the blasting cap. Initiate the sample in accordance with 4.4.1. Verify the complete detonation of the sample's blasting cap. Verify the dent block measurement. The sample(s) and lot shall be rejected if any blasting cap does not function completely or if any dent block measurement does not meet the requirement.

4.6 Packaging inspections.

4.6.1 Packaging material moisture content. At the start of each production day, three (3) samples of all paper packaging material contained within the barrier bag shall be randomly selected and subjected to moisture content analysis. Not less than ten (10) grams of each sample shall be accurately weighed and placed in a tared weighing dish. Each dish and contents shall be weighed and placed in an oven and dried at 100 ± 2°C for two (2) hours. Each dish and contents shall be cooled in a desiccator for not less than fifteen (15) minutes and weighed. The loss in weight shall be calculated as percent moisture in each sample. If any sample(s) does not meet the moisture requirement, the sample(s) and that production day's packaging material shall be rejected.

4.6.2 Leakage test. The test shall be performed on the sealed barrier bag and its contents (actual or simulated) in accordance with MIL-STD-2073-1 Leakage Test paragraph G.4.2. The test shall follow either the vacuum retention, the submersion (or immersion), or the hot water technique. If any sample does not meet the leakage test, any packaged sample(s) in that shift shall be rejected.

4.6.3 Heat seal seam strength test. The test shall be performed on the sealed barrier bag in accordance with MIL-STD-2073-1 Heat seal seam strength test paragraph G.4.3. The samples shall be chosen from the final heat seal seam which closes/seals the pre-assembled bag. If any sample does not meet the heat seal seam test, any packaged sample(s) in that shift shall be rejected.

4.6.4 Metallic seal inspection. The metallic seal shall be inspected for presence, proper markings, and correctness of application. Pull on the metallic seal to determine if it is secure. If the seal breaks the sample and item lot shall be rejected. If the wire breaks before the seal, the box shall be resealed using a new metallic seal.

5. PACKAGING

5.1 Packaging and marking. The materials, components, and assemblies shall comply with all the requirements of the drawings listed in 13023563 and all its subsidiary drawings.

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5.1.1 Barrier bag resistance to leakage. The barrier bag shall not leak as evidenced by performing MIL-STD-2073-1 Para G.4.2 (leakage test) and MIL-STD-2073-1 Para G.4.3 (heat-sealed seam test).

5.1.2 Marking ink/media. The ink and/or marking media shall be in accordance with A-A-208 and/or A-A-59383. All markings shall be smooth, uniform, clear in appearance and free of conspicuously ragged edges. There shall be no bleeding of color markings or the plastic exterior to the extent that there is a mottled or streaky appearance. The markings shall show no lifting, wrinkling, or other visible effect on the plastic's surface. The markings shall be applied on clean, dry surfaces that shall be free of dirt, oil, grease, water, corrosion, or other foreign matter. All required markings shall be present, neat and legible, only where specified, and sharply defined.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M14 is part of the family of Modernized Demolition Initiators (MDI). The M14 is used to initiate all past and current fielded and developmental demolition items such as explosives, shaped charges, and cratering charges. The burn time will increase with altitude and colder temperatures. The M14 has been designed to allow a minimum 5-minute delay, under all weather and altitude conditions, to allow personnel to move to the minimum safe distance from the explosive charges being detonated.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2 and applicable documents from Section 6.
b. Requirement/CDRL for Physical Configuration Audit (PCA).
c. Requirement/CDRL for First article inspection.
d. Requirement/CDRL for Conformance inspection.
e. Requirement/CDRL for Ammunition Data Cards (ADC) as specified in MIL-STD-1168.
f. Requirement/CDRL for Propellants and Explosives data sheet.
g. Requirement/CDRL for Acceptance Inspection Equipment (AIE) and associated test procedures. M81's used during LAT should be submitted as part of the AIE.
h. Requirement/CDRL for Alternate Acceptance Plans.
i. Packaging requirements. Special requirements exist for the packaging of this item. Detailed directions and additional information may be obtained from FCDD-ACE-LP.

6.3 Burn rate conversion. For burn time testing, hot temperature sample(s) recorded burn times should be adjusted to sea level using the following formula:

Adjusted burn time = Recorded burn time – [(Elevation in feet / 3772 ft) x 5.5 seconds)]

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It is recommended that the adjusted burn time be tested per lot of time fuse prior to marking the fuse in order to calibrate burn time.

6.4 High order detonation. The complete detonation of the M14 is evidenced by the propagation of the time fuse which initiates the blasting cap. The blasting cap detonates high order, fully consuming all explosive material

6.5 Critical characteristic. Inadvertent initiation of the blasting cap can lead to injury and death.

a. The blasting cap cup being split or cracked, explosive charge being exposed, or there being evidence of explosive leakage is a critical characteristic. The explosives contained within the blasting cap should be contained within the blasting cap cup. Lead Azide is highly sensitive to impact and static discharge. The cup should not be split or cracked as this may lead to energetic leakage and these defects can lead to injury and death during handling and usage.

b. The burn time of any time increment being less than 50% of the desired outcome is a critical characteristic. Based on historical evidence that during burning the time fuse could "jump" and reduce the burn time significantly which would not allow safe exit outside the surface danger zone of the main explosive charge.

6.6 Document history. This document supersedes QAA-1424J Performance Specification for the Cap, Blasting: Non-Electric , Delay, M14.

6.7 Subject term (key word) listing.
MDI (Modernized Demolition Initiators)
Timed demolitions

6.8 Overview of changes. ECP R20Q2003 provides documentation for noted changes.

Custodian:
Army – AR
Navy – MC
Air Force – 170

Preparing activity:
Army – AR
(Project: 1375-2020-003)

Review activities:
Navy – AS, SH
Air Force – 11, 20

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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